Prathamesh Patil

+91 73490 41840 | Bengaluru, Karnataka
prathameshpatil330@gmail.com
LinkedIn | Github

PROFESSIONAL SUMMARY

Final-year AI/ML Engineering student with hands-on experience building production-grade RAG pipelines, multi-agent system features, and ML-backed REST APIs. Demonstrates curiosity, discipline, and ownership in building AI system integration. Proficient in Python, FastAPI, and cloud-native deployment. Published two research papers; Salesforce Agent Development certified.

TECHNICAL SKILLS

AI ML Development: Python, TensorFlow, scikit-learn, PyTorch, RAG Pipelines, FastAPI, Unit Testing
Web Development: Node.js, Express.js, React.js, Next.js, REST APIs, WebSockets, Agile, Scrum
Databases and Infra: SQL, PostgreSQL, Redis, Docker, Kubernetes, AWS, CI/CD
Core CS: Data Structures, Algorithms, OOP, Operating Systems, DBMS
Tools: Git, LLM APIs
Additional Skills: React

WORK EXPERIENCE

Software Engineer Intern - AI Division, *Jmedia Corp* — Jan 2025 – Jun 2025

- Designed and deployed scalable ML-backed REST APIs using FastAPI and AWS Lambda, integrating LLM APIs and OOP design patterns with rate-limiting middleware to reduce response latency by 35% and improve SLA compliance.
- Containerized enterprise microservices with Docker and orchestrated via Kubernetes, automating canary deployments through Git-based GitHub Actions CI/CD to cut release cycle time by 60%; secured services with JWT auth and gRPC routing.
- Instrumented robust services with structured event logging and Prometheus metrics; built weekly React dashboards in Grafana to track p95/p99 latency, error rates, and throughput for engineering and product stakeholders.

AI & Cloud Computing Intern, *SuprMentr Technology Pvt Ltd* — Nov 2025 – Mar 2026

- Engineered an adaptive multi-agent RAG platform leveraging LangGraph and ChromaDB, integrating LLM APIs for self-evaluation workflows that tracked hallucination rate and accuracy, enabling iterative error reduction via robust self-correction loops.
- Developed complex SQL queries (window functions, CTEs, aggregations) on PostgreSQL/pgvector and visualized weekly usage reports using React dashboards, supporting data-driven product decisions with scalable analytics.
- Integrated Redis Streams for event-driven messaging between agent nodes, utilizing Git version control and Groq Llama 3.3 plus Gemini embeddings to reduce infrastructure costs by ~30% while maintaining high-quality response performance.

PROJECTS

Adaptive Agentic RAG - Multi-Agent Research System

- Architected a LangGraph multi-agent pipeline with conditional routing between local vector stores and real-time web search, using OOP-based self-correction nodes for hallucination detection, coordinated via Kafka.
- Designed persistent semantic user profiling with PostgreSQL + pgvector, combining Groq Llama 3.3 for low-latency reasoning with 3072-dimension Gemini embeddings to reduce API costs.
- Built a React + FastAPI frontend deployed on AWS with OpenTelemetry instrumentation for full-stack production observability.
- **Tools Used:** Python LangGraph Groq (Llama 3.3) PostgreSQL + pgvector ChromaDB FastAPI

AI-SaaS-Kit - Scalable Multimodal AI Boilerplate

- Built an asynchronous multimodal AI pipeline using Gemini 1.5 Flash for VLM reasoning, orchestrated through Celery workers and Redis caching, applying an OOP repository pattern for data access.
- Implemented secure, type-safe data flow via JWT-authenticated proxy routes with rate limiting and strict Pydantic v2 serialization.
- Operationalised CI/CD with GitHub Actions (automated testing, multi-stage Docker builds) and Kubernetes-ready manifests; enabled canary deployments on AWS with real-time WebSocket and gRPC event synchronisation.
- **Tools Used:** Next.js FastAPI PostgreSQL Redis Celery WebSockets gRPC Docker

EDUCATION

Dayananda Sagar Academy of Technology and Management, Bengaluru, B.E. in Artificial Intelligence & Machine Learning 2022 – 2026